

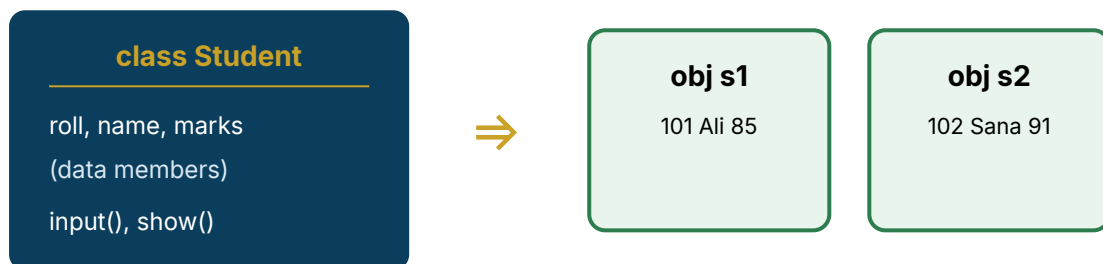
OOP in C++: class, object, constructor, inheritance, polymorphism

★ Golden

One college period · 40-45 minutes · OOP using C++ and file handling

What you will learn

- State encapsulation, inheritance, polymorphism
- Write a simple class Student with constructor
- Explain public, private, protected
- Open text/binary files for read, write, append



A class is a blueprint; objects are instances with their own data.

- Encapsulation: wrap data + functions; hide data with private.
- Inheritance: derived class reuses base class (reusability).
- Polymorphism: one name, many forms — overload (same name, different parameters) and override (same signature in derived class).

```
#include <iostream>
using namespace std;
class Student {
    int age; float percentage;           // private by default
public:
    Student() { age = 0; percentage = 0; }           // default constructor
    Student(int a, float p) { age = a; percentage = p; } // overloaded
    void show() { cout << age << " " << percentage; }
    ~Student() { } // destructor
};
int main() {
    Student s1(17, 82.5);
    s1.show();
}
```

Access specifiers: private (only inside class), public (everywhere), protected (class + derived).

Constructor has the class name, no return type, called automatically. Destructor ~ClassName() at end of life.

Inheritance sketch

```
class Base { public: int x; };
class Drive : public Base { public: int y; };
// Drive object can use x and y
```

File handling

Text files store characters; binary files store raw bytes (objects). Modes: ios::in read, ios::out write (truncate), ios::app append. BOF beginning, EOF end. Streams: get/put (char), getline (string), read/write (binary).

```
#include <fstream>
ofstream out("data.txt");
out << "Ali 80\n";
out.close();
ifstream in("data.txt");
string line; getline(in, line);
```

HOW BIEK ASKS THIS

Sindh lab: class STUDENT with AGE and PERCENTAGE; TIME class with constructors; inherit DRIVE from BASE; write/read text file; write/read student object as binary.

اردو میں سمجھیں

کلاس خاکہ ہے، آبجیکٹ اس کی کاپی۔ Encapsulation ڈیٹا چھپاتا ہے، Inheritance وراثت، Polymorphism ایک نام کئی کام۔
Constructor آبجیکٹ بنتے وقت چلتا ہے۔
فائل بینڈلنگ: ٹیکسٹ اور بائنری، read/write/append.

CHECK YOURSELF

1. Does a constructor return a value?
2. ios::app means?

Answers: 1. No — not even void · 2. Append to the file without erasing old data

MCQs

1. Wrapping data and functions together is:
 - a. inheritance
 - b. encapsulation
 - c. compilation
 - d. paging
2. A constructor has:
 - a. the class name and no return type
 - b. any name
 - c. return type int
 - d. a destructor symbol

Key: 1=b, 2=a

Short questions (3 marks each)

1. Explain class and object.
2. Differentiate constructor and destructor.
3. Differentiate text and binary files.

Model short answers

1. Class: user-defined type with data members and member functions. Object: variable of that type.
2. Constructor initialises (same name, auto-called). Destructor cleans up (~Name), no arguments.
3. Text: human-readable characters. Binary: exact memory image, efficient for objects.

Long questions (board Section C)

1. Explain OOP principles with a C++ class that uses a constructor. Discuss inheritance and polymorphism briefly.